

# PROJECT QUANT — GVM DEVELOPMENT MEMORY

G-Series | Build Journey G1 to G7 | Platform Architecture Locked

Arpit Goel + Claude | April-May 2026

## CORE MISSION

93% of retail investors in India lose money — not because they lack intelligence, but because they lack the tools institutions have. Project Quant exists to change that.

Build AI Aladdin — India's first institutional-grade stock intelligence platform for retail investors. A platform where a 25-year-old investor in Lucknow has the same analytical edge as a fund manager in Mumbai.

## BUILD STATUS — G1 TO G7

| Session | Date        | Key Achievement                                                                                                  | Status |
|---------|-------------|------------------------------------------------------------------------------------------------------------------|--------|
| G1      | Apr 2026    | Project vision locked. Tech stack decided. Python + VS Code install planned.                                     | Done   |
| G2      | Apr 2026    | Python 3.14 + VS Code + Pandas installed. gvm_scorer.py created. First GVM skeleton written.                     | Done   |
| G3      | Apr 2026    | gvm_engine.py built. screener_loader.py reading CSV. All 21 parameters mapped.                                   | Done   |
| G4      | 30 Apr 2026 | 6 logic fixes applied. 1,806 stocks scored. Input 2 Final — 127 segments completed.                              | Done   |
| G5      | 7 May 2026  | 3,371 merge issue found. PostgreSQL setup next. Pending duplicate fix + validation.                              | Done   |
| G6      | 19 May 2026 | Platform architecture expanded. Internal EXE track added. API Library defined. Delivery model locked.            | Done   |
| G7      | 20 May 2026 | Railway-first architecture locked. No MCP. No Cowork. No Desktop dependency. <b>Models in 6 months via chat.</b> | Done   |

## TECH STACK — LOCKED

| Component        | Technology                      | Status                 |
|------------------|---------------------------------|------------------------|
| Backend          | Python + FastAPI                | Building               |
| Frontend         | React                           | Week 3                 |
| Database         | PostgreSQL on Railway           | G7 next task           |
| Hosting Backend  | Railway (free tier)             | Pending deploy         |
| Hosting Frontend | Vercel (free tier)              | Pending deploy         |
| Data Source      | Screener.in CSV + Input 2 sheet | Live                   |
| Historical Data  | Yahoo Finance (yahoo_hist.py)   | Live — no MCP needed   |
| Live Quotes      | Yahoo Finance (yahoo_quote.py)  | Live — no token needed |
| AI Layer         | Claude API                      | Week 4                 |
| Version Control  | GitHub                          | Pending setup          |
| Sheet Reading    | gspread + service account       | Installed              |
| MCP              | REMOVED — not needed            | Deleted                |
| Fyers API        | REMOVED permanently             | Deleted                |
| Claude Desktop   | Not required                    | Dropped                |

## RAILWAY-FIRST ARCHITECTURE — LOCKED IN G7

Decision made May 20, 2026: No MCP. No Cowork. No Desktop dependency. Railway is the brain. Browser is the interface.

| Layer | What it does | Status |
|-------|--------------|--------|
|-------|--------------|--------|

|                       |                                                 |                    |
|-----------------------|-------------------------------------------------|--------------------|
| PostgreSQL on Railway | Stores all GVM scores, price history, signals   | G7 next task       |
| FastAPI on Railway    | Exposes all data as API endpoints — always on   | Week 2             |
| Browser Claude        | Fetches Railway URLs — reads live data anywhere | After Railway live |
| VS Code               | Write and deploy code only — no data dependency | Active             |
| Google Sheets         | Source data via gspread — V5 live prices        | Active             |

### How Claude reads data without MCP:

- FastAPI deployed on Railway — public URL, API key protected
- Browser Claude calls web\_fetch on Railway URL — gets GVM score, signals, prices
- Works from mobile, laptop, desktop, anywhere with internet
- No VS Code terminal running. No node process. No token refresh.
- PC can be off — Railway runs 24/7

## 6 MODELS IN 6 MONTHS — BUILD PLAN

Pivot decision May 20, 2026: Build 6 models via chat (not 12, not Cowork). Quality over speed. Chat = deep understanding + zero silent bugs.

| # | Model             | Why Priority                            | Timeline  |
|---|-------------------|-----------------------------------------|-----------|
| 1 | GVM Rating Engine | Core product — already 70% built        | Month 1-2 |
| 2 | V5 Signal Engine  | Live trading — real P&L impact daily    | Month 2-3 |
| 3 | Portfolio Health  | Client money — Rs.60L V4 capital        | Month 3-4 |
| 4 | Screener Engine   | Feeds GVM + V5 — daily use              | Month 4   |
| 5 | Alert System      | Makes V5 autonomous during market hours | Month 5   |
| 6 | Pair Trading      | Highest alpha — proven edge             | Month 6   |

### Why chat over Cowork for model logic:

- Same code quality — zero difference in output
- Chat catches bugs immediately — Cowork may build them in silently
- Financial product — wrong GVM score = wrong investment decision = user loses money
- After chat build, Arpit owns every layer and can direct any developer confidently
- Cowork revisit at Month 4-6 for UI, reports, formatting only — never model logic

## GVM MODEL ARCHITECTURE

### G — Growth (14 Parameters)

Operational (9): 5Y Sales Growth, 3Y Sales Growth, 5Y Profit Growth, 3Y Profit Growth, QoQ Sales Growth, QoQ Profit Growth, OPM, OPM Expansion, Fixed Asset Growth

Reliability (5): Promoter Holding, Institutional Holding Change, ROCE, Interest Coverage, Dividend Yield

### V — Value (2 Parameters)

PE vs Peer Group (inverted logic — lower PE = higher score) | Potential Annual Upside

### M — Momentum (5 Parameters)

1Y Return, 3Y Return, DMA50, DMA200, 52-Week vs Index

### Scoring Logic

2-Factor System: Factor 1 (absolute performance) + Factor 2 (relative vs peer group average), averaged. Scale: 2.5 / 5 / 7.5 / 10. Absolute thresholds: >15% = 10, 5-15% = 7.5, 0-5% = 5, below 0% = 2.5. PE uses inverted logic. Peer averages filtered to exclude outliers. GVM = G + V + M combined.

G7 CHECKLIST — ACTIVE

| Priority | Task                                                             | File                           | Status  |
|----------|------------------------------------------------------------------|--------------------------------|---------|
| 1        | Fix duplicate merge (3,371 to 1,971)                             | screener_loader.py             | Pending |
| 2        | Fix segment count (156 to 127)                                   | input.csv / screener_loader.py | Pending |
| 3        | Fix Stylam segment — Building Materials                          | input.csv                      | Pending |
| 4        | Validate 5 stocks: HDFCBANK, SUNPHARMA, TATASIGNAL, INFY, MARUTI | gvm_engine.py                  | Pending |
| 5        | PostgreSQL on Railway: quant_db, 4 tables, Day1_bands in (test)  | db_handler.py                  | Pending |
| 6        | GitHub setup + first commit — all project files                  | All files                      | Pending |
| 7        | FastAPI GVM Score API — single endpoint live on Railway          | main.py                        | Pending |
| 8        | Claude browser fetches Railway URL — confirm browser test        | browser_test                   | Pending |

QUANT\_PROJECT FOLDER — FILE STATE

| File                 | Purpose                                    | Status                 |
|----------------------|--------------------------------------------|------------------------|
| gvm_engine.py        | Core scoring logic — all 21 parameters     | v3 — G4 fixes applied  |
| screener_loader.py   | Reads CSV, cleans, merges, calls engine    | v3 — merge bug pending |
| input.csv            | Input 2 Final — 1,971 stocks, 127 segments | Live                   |
| screener.csv         | Screener.in export — 3,222 stocks          | Latest                 |
| yahoo_hist.py        | Yahoo historical OHLC — DMA, returns       | Live — no token needed |
| yahoo_quote.py       | Yahoo live quote — 15 min delay            | Live — created May 20  |
| service_account.json | Google Sheets service account              | Live                   |
| main.py              | FastAPI app — to be built                  | Pending G7             |
| db_handler.py        | PostgreSQL handler                         | Not created yet        |
| fyers_* files        | ALL DELETED permanently May 20             | Gone                   |
| yahoo_mcp.js         | Yahoo MCP server — built but deprioritised | Optional               |

DATA SOURCE RULE — NON-NEGOTIABLE

Yahoo Finance (yahoo\_quote.py + yahoo\_hist.py) = ALL price data. Historical daily and near-live quotes. No token. No auth. Always works.

Fyers = PERMANENTLY DELETED. Never reinstall. Never reference.

MCP = DROPPED. Railway FastAPI replaces MCP permanently.

GROUND RULES

| Rule                                  | Detail                                                                   |
|---------------------------------------|--------------------------------------------------------------------------|
| No PostgreSQL until validation passes | 5 test stocks must score correctly before DB insert                      |
| Railway-first                         | All data on Railway PostgreSQL — accessible from any device, any browser |
| No MCP ever again                     | FastAPI on Railway replaces MCP permanently                              |
| No Fyers ever again                   | Yahoo Finance handles all price data — free, no token                    |
| Blank values = 5                      | Missing data defaults to neutral score                                   |
| Interest Coverage — skip BFSI         | Banks/NBFCs excluded from this parameter                                 |
| Segment min 5 / max ~25 stocks        | 127 GVM segments                                                         |
| Learning rule                         | Explain WHY before HOW. Arpit directs. Claude executes.                  |

|                      |                                                           |
|----------------------|-----------------------------------------------------------|
| 6 models in 6 months | Chat only. No Cowork for model logic. Quality over speed. |
|----------------------|-----------------------------------------------------------|

## NEXT 10 TASKS — SEQUENCED ROADMAP

| #  | Task                                                                     | Timeline       | Depends On                 |
|----|--------------------------------------------------------------------------|----------------|----------------------------|
| 1  | Fix merge + segment + Stylam. Validate 5 stocks.                         | This week (PC) | quant_project folder on PC |
| 2  | GitHub: Push entire quant_project.                                       | Same as G7     | G7 complete                |
| 3  | PostgreSQL on Railway: quant_db live. 4 tables. Day 7 continues started. | Week 2         | 5-stock validation passed  |
| 4  | FastAPI on Railway: GVM Score API live. Claude fetches from browser.     | Week 2         | PostgreSQL on Railway      |
| 5  | V5 Data Layer: Yahoo indicators + GVM segments via FastAPI.              | Week 2         | GVM API live               |
| 6  | V5 Signal Engine in Python — Buy Reversal + Sell RSI                     | Month 2        | V5 data layer              |
| 7  | Portfolio Health API — holdings to GVM health score                      | Month 2        | GVM API live               |
| 8  | Screener APIs x2 — Buy Reversal + Sell RSI as endpoints                  | Month 3        | Portfolio API              |
| 9  | Alert System — signal triggers Telegram/email.                           | Month 3-4      | Screener APIs              |
| 10 | Claude AI Explanation Layer — plain English for every GVM score.         | Month 4        | All core APIs              |

G Memory — GVM Development | Updated May 20, 2026

Next session command: Open VS Code on PC — fix screener\_loader.py merge issue — setup PostgreSQL on Railway — push to GitHub.